A Randomized Clinical Trial Comparing The Overall Adverse Event Rate Of Inguinal Hernia (IH) Repair Prior To NICU Discharge Versus IH Repair After NICU Discharge And Beyond 55 Weeks Post Menstrual Age In Premature Infants.

Short Title: Timing of Inguinal Hernia Repair in Premature Infants: A Randomized Trial

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HIP Trial (**H**ernia **i**n **P**remies)

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Abstract

Very few commonly used neonatal surgical therapies have been evaluated to determine their safety. Premature infants are especially vulnerable to unrecognized treatment hazards. Inguinal hernia (IH) repair is the most commonly performed major operation in premature infants, is associated with a 30-40% morbidity rate, and it is unknown whether the operation should occur prior to neonatal intensive care unit (NICU) discharge or be delayed. Early IH repair likely increases anesthesia and cardiorespiratory serious adverse events (SAEs), whereas delayed IH repair likely increases hernia-related adverse events. A growing concern is the possibility that early anesthesia exposure impairs neurodevelopment. The randomized clinical trial (RCT) proposed will be the first trial to evaluate the effect of timing and duration of anesthesia exposure on neurodevelopmental outcomes. Our objective is to conduct a multicenter RCT to determine whether early inguinal hernia repair (prior to NICU discharge) or late inguinal hernia repair (~5 months after NICU discharge) is the safer surgical approach for premature infants who are diagnosed with an inguinal hernia while in the NICU. Assuming a 60% randomization rate, which is supported from a feasibility pilot RCT, 615 infants will be randomly assigned to early or late IH repair over a 2.5-year period. The primary hypothesis is that late IH repair is safer as defined by a 10% absolute reduction in the proportion of infants with $\geq$1 SAE (relative risk = 0.66; number needed to harm of 10 for early repair) within 10 months after enrollment. The primary outcome, as adjudicated by a committee masked to treatment group assignment, is the proportion of infants with $\geq$1 SAE during the study period. Secondary outcome measures include the number of hospital days during the study period (hypothesis: median 3 days less with late IH repair) and, in selected centers (n=200 infants), neurodevelopmental testing with the Bayley Scales of Infant Development (BSID), 3rd edition, at 2 years corrected age (hypothesis: 0.4SD higher Cognitive Composite score with late IH repair). The potential differences in treatment effect according to prespecified patient characteristics (at enrollment) will be examined by frequentist and conservative Bayesian methods and a method to predict the relative safety of early and late IH report for individual infants will be developed and tested. The trial is utilizing the Pediatric Surgery Research Collaborative (PedSRC), a recently established cooperative group committed to performing prospective clinical studies in pediatric surgery. Successful completion of the RCT proposed will stimulate future studies within pediatric surgery.

Statement of Problem

The problem we are addressing in the HIP Trial is the lack of knowledge regarding whether IH repair in premature infants who are diagnosed while in the NICU is more safely performed, for the majority of infants, prior to discharge (early repair) or ~5 months after NICU discharge (late repair). Additionally, the timing (and duration) of general anesthesia is believed to be an important determinant of neurodevelopmental outcomes beyond infancy. The HIP trial will also compare neurodevelopmental outcomes, as assessed by the Bayley Scales of Infant Development, 3rd edition (BSIDIII), at 2 years corrected age, for infants in the early versus late IH repair groups.

Hypotheses

<u>Primary</u>: Late IH repair (5 months after NICU discharge, ~ 55-60 weeks post-menstrual age [PMA]) in premature neonates with an IH will result in a 10% absolute reduction in the proportion of infants with $\geq$ 1 SAE compared to early IH repair (prior to NICU discharge, ~ 38 weeks PMA).

<u>Secondary</u>:

- Late IH repair will be associated with a 0.4 SD increase in mean scores on the BSIDIII Cognitive Composite score (6 points absolute difference) compared to early IH repair.
- Despite an *overall* treatment effect favoring late IH repair, the actual treatment effect will vary for pre-defined and specified patient subgroups.
- Successful conduct of this RCT, as defined by $\geq$60% randomization and achievement of required sample size, will further establish the Ped SRC as an entity that will promote further conduct of prospective clinical trials within pediatric surgery and foster successful grant submissions by junior and established pediatric surgeons.

Specific Aims

Primary Aim: Compare the overall safety of early and late IH repair with respect to significant adverse events (SAEs) and total hospital days in the 10 months after randomization.

The **primary outcome measure** is the proportion of infants with $\geq$1 SAE, measured from enrollment through 10 months after enrollment. Final determination of SAE will be by an independent adjudication committee, masked to treatment group assignment, consisting of an experienced pediatric surgeon, a neonatologist, and others, at their discretion.

Secondary outcome measures will include the number of hospital days from enrollment through 10 months beyond enrollment. This measure was chosen because it is easily measured, objective, and is a direct measure of the impact of underlying SAEs. With regard to this outcome, we hypothesize that late IH repair will be associated with a 3 day reduction in the median total number of hospital days during this period (18 hospital days for early IH repair versus 15 for late IH repair). Additionally, participants will be asked to return to Vanderbilt's neurodevelopmental clinic for a neurodevelopmental follow-up exam when they are 22-26 months corrected age (chronological age - # weeks/months premature). A standard neurological exam and the Bayley Scales of Infant Development, 3rd Edition (BSIDIII), will be administered by a trained examiner at this follow-up visit. The BSIDIII is a standardized assessment and was chosen for its validity, reliability, ease of administration, and widespread use for the targeted age group across facilities.

Secondary Aims:

1. Compare the overall safety of early and late IH repair with respect to neurodevelopmental outcome at 22-26 months corrected age (at selected sites with well-established follow up programs).
2. Using Bayesian and frequentist analyses, evaluate specific hypotheses about patient characteristics that may influence the relative safety of early and late IH repair.
3. To conduct the HIP Trial in an exemplary manner and thereby promote the newly established pediatric Surgery Research Collaborative as an ongoing research network.

Rationale / justification

Lack Of Neonatal Surgery Safety Studies. The overall problem we are addressing is the near total lack of well-designed research evaluating the safety of neonatal surgical therapies. In part due to the lack of evidence, current decision-making regarding neonatal surgery relies primarily on the preferences of surgeons and neonatologists. Although robust efforts have been made in many fields to assess and improve patient safety since the Institute of Medicine report "To Err is Human", the same cannot be said for neonatal surgery.[1-2] There is a critical void in our knowledge regarding the safety of even the most common neonatal surgical therapies.[3-4] Decades of experience in neonatal medicine demonstrate the dangers of using therapies without rigorous testing of their short and longer-term safety.[5-11] This knowledge gap in one of the most vulnerable of all patient populations is unacceptable and is the focus of our proposal.

Timing Of Premature Inguinal Hernia Repair. We have chosen to focus on IH repair in premature infants because: 1) It is the most commonly performed major operation in premature infants[12,13]; 2) The morbidity rate is 30-40% in this vulnerable population[14-17]; 3) Current care varies and is nearly equally split between early IH repair (prior to NICU discharge, at ~38 weeks PMA) and late IH repair (5 months after NICU discharge, at ~ 55-60 weeks PMA)[18,19]; 4) Physician treatment preferences are not insurmountable (unlike with some surgical treatments), the recruited surgeons are willing to randomize infants to receive either therapy, and parental consent would be much more easily obtained than in an RCT of emergency surgical procedures in unstable infants; and 5) A RCT of IH repair in premature infants offers a unique opportunity to study the effect of timing of general anesthesia on neurodevelopmental outcomes. The American Academy of Pediatrics Committee on Fetus and Newborn and the Section on Surgery recently highlighted the surgical care of premature infants with IH as problematic, associated with high morbidity and variable care, and needing multicenter randomized trials to compare the treatment options.[20]

Unlike IH repair in term infants or older children, IH repair in premature infants is a technically demanding operation with many potential operative and anesthetic risks. Early IH repair is associated with a high risk of cardiorespiratory and anesthetic adverse events (AEs) and potentially an increased risk of intra operative technical AEs (e.g. injury to vas deferens, gonadal vessels, intestine, or other structures).[21-24] However, the longer IH repair is deferred, the longer the infant is exposed to potential hernia-related AEs (e.g. incarceration, strangulation, unplanned readmission, emergency operation), although cardiorespiratory and anesthetic AEs are likely decreased.[25-26] The controversial clinical question involving this surgical choice (early v. late IH repair) is almost entirely focused on safety, rather than effectiveness.[20] Despite many retrospective case series, there has never been a multicenter RCT to adequately address the very fundamental question of whether IH repair should be performed prior to NICU discharge or should be deferred until a later time in premature infants. There is a large gap in current knowledge regarding the relative safety of each surgical approach.[20]

The RCT we propose--the HIP Trial (Hernia in Premies Trial)--is a true safety study with the goal of measuring and reducing overall serious adverse events, “resulting from a medical intervention, or lack thereof”. Measuring and reducing safety, or harm, is difficult although there are multiple methods available.[27] We will measure serious adverse events (SAEs), defined as an unintended injury or complication caused by health care management resulting in an escalation of care, disability at discharge, death, prolonged stay in hospital or a subsequent admission to the hospital. Health care management includes the decisions and actions of individual providers (e.g. whether to perform IH repair prior to NICU discharge, or withhold such treatment until later) in addition to the broader systems and process of care. These definitions are evidence-based and adapted from existing safety literature.[28,29] Although patient safety definitions vary, the universal goal of any patient safety effort is to minimize the incidence and impact of, and maximize recovery from, adverse events.[30] “Patients who fail to receive needed treatments, or who are subjected to the risks of unneeded care, are placed at risk for injury every bit as objectionable as direct harm from a surgical mishap.”[1] In addition to carefully measuring important patient-centered safety outcomes, we are doing this in the context of a RCT, which is uncommon and highly desirable within the safety discipline. Adverse outcomes that are caused by the treatment decisions and actions of individual providers are best distinguished from those resulting from the underlying patient problems by using the study design we propose: a pragmatic RCT. If one surgical strategy is shown to be associated with fewer SAEs (as hypothesized), the pragmatic RCT design also offers the best chance of changing provider behavior towards adopting the safest approach.[31]

Anesthesia Risk. The importance of the HIP trial relates partly to the serious and growing concern that anesthesia during infancy[32,33] may have long-term adverse effects on the developing brain, particularly in premature infants who have other insults.[34] The vast majority of premature infants undergoing IH repair have general anesthesia. Multiple studies have identified an association between general anesthetic and neurotoxicity[35-39], however, none of these studies were randomized clinical trials and have serious limitations as such. The bulk of information on the detrimental effects of anesthesia exposure comes from animal models and the duration of anesthesia exposure in these studies is typically in excess of what is typical in pediatric surgical procedures[33]. Inguinal hernia repair is considered the standard and required treatment for infants with an inguinal hernia. Anesthesia (general, regional, or both) is a necessary component of this surgical treatment.. Further study is critical to define the window of vulnerability, identify the extent of anesthesia-induced neuronal alterations and the functional end points, and assess how to avoid the potential adverse effects. [40] The FDA’s Anesthetic and Life Support Drugs Advisory Committee concluded that the evidence was strong enough to forego elective procedures in children less than 3 *years* old.[33] This report also goes on to say that due to a lack of sufficient evidence, an association between anesthesia exposure and neurodevelopmental disabilities cannot be made. Furthermore, the FDA urges the scientific and medical communities to produce additional research, specifically randomized clinical trials, to fully understand what implications, if any, result from infant exposure to anesthesia.
Whether general anesthesia causes adverse neurodevelopmental outcomes is very difficult to determine in humans, in part because of multiple confounders in observational studies comparing exposed and unexposed children. There is only one relevant ongoing RCT (and none previously).[41] In this trial (GAS Trial, Clinicaltrials.gov NCT00756600), infants with an IH are randomized to either general or regional anesthesia. The primary outcome is the Wechsler Preschool and Primary Scale of Intelligence full scale IQ score at 5

years. Limitations of this trial in assessing effects on patient safety is that it may not be very generalizable to institutions that have limited expertise with regional anesthesia in infants and SAEs that may occur with regional but not general anesthesia (e.g., intraoperative adverse events with patient movement under regional anesthesia only) are apparently not being measured.

Our proposal addresses what we regard to be a fundamental flaw in the current research agenda related to the risk of anesthesia in infants -- the lack of surgeon involvement and leadership. Surgeons are key decision makers regarding the necessity of interventions that require anesthesia and the timing of these procedures. Early IH repair is likely to entail not only earlier exposure but also longer anesthesia times (due to increased technical difficulty). The HIP trial offers a unique opportunity to evaluate what parents and providers care about most, i.e. the combined, “real world” effects of surgery and anesthesia on the outcomes of vulnerable premature infants, to be identified in a large pragmatic RCT that avoids the confounding that plagues observational studies.

Background / Previous studies

Dr. Blakely previously performed a small pilot RCT addressing the timing of IH repair in premature infants.[14] Early IH repair was compared to late repair at a goal PMA of 55 weeks; 39 of 50 eligible infants at 4 participating sites were randomized (78% randomization rate).

Table 1. Reduction in Serious Adverse Events and Hospital Stay with Early IH Repair in Pilot RCT.

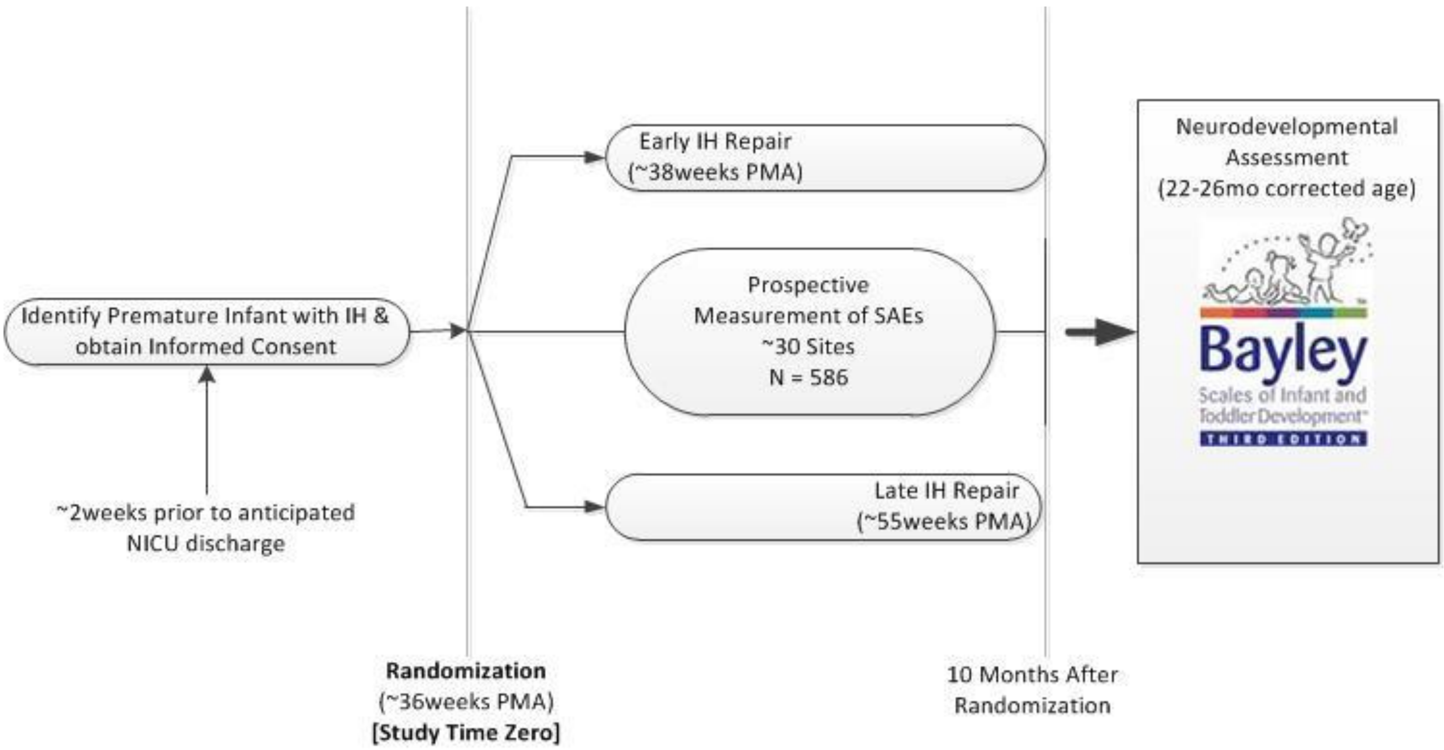
Characteristic of Randomized Infants / Outcome	Early IH Repair N = 19	Late Repair N = 20*
Estimated gestational age (weeks)	28.9 (24 – 32)	28.4 (25 – 34)
Birth weight (grams)	959 (520 – 1,550)	958 (500 – 2,710)
PMA at operation (weeks)	38.4 (34 – 49)	48.6 (33 – 61)**
Weight at operation (kilograms)	2.55 (1.8 – 4.0)	3.9 (1.3 – 8.9)
Post operative apnea or bradycardia	4/19 (21%)	0
Reintubation or failure to extubate at end of operation	5/19 (26%)	2/12 (17%)
Any anesthetic adverse event	9/19 (47%)	2/12 (17%)
Post operative length of stay (median, range)	8 (1-206)	1 (0-30)
Medical attention required for IH reduction	0	6/12 (50%)
Death after IH repair	0	0

*IH repair was not performed during the study period in 8 infants in the late IH group: 2 died from complications of prematurity after NICU discharge, 5 were lost to follow-up, and one had no identifiable IH at follow up exam. **6 infants had repair before 55 weeks PMA (range: 33-49 weeks) due to hernia incarceration (n=4) or parental/provider preference (n=2). While intra- and postoperative SAEs were accurately assessed, a limitation of this pilot study was the loss to follow-up of 5 infants and the lack of measurement of all hospital days during the study period, rather than postoperative hospital days only.

Methods

To test our hypotheses, multiple Participating Clinical Centers, including a Clinical Coordinating Center (Vanderbilt), and a Data Coordinating Center (Vanderbilt) will conduct this multi-center, randomized clinical trial. The acronym for the trial is the HIP trial (Hernia in Premies) and it has been registered with Clinicaltrials.gov (NCT01678638).

Figure 1. HIP Trial Study Design

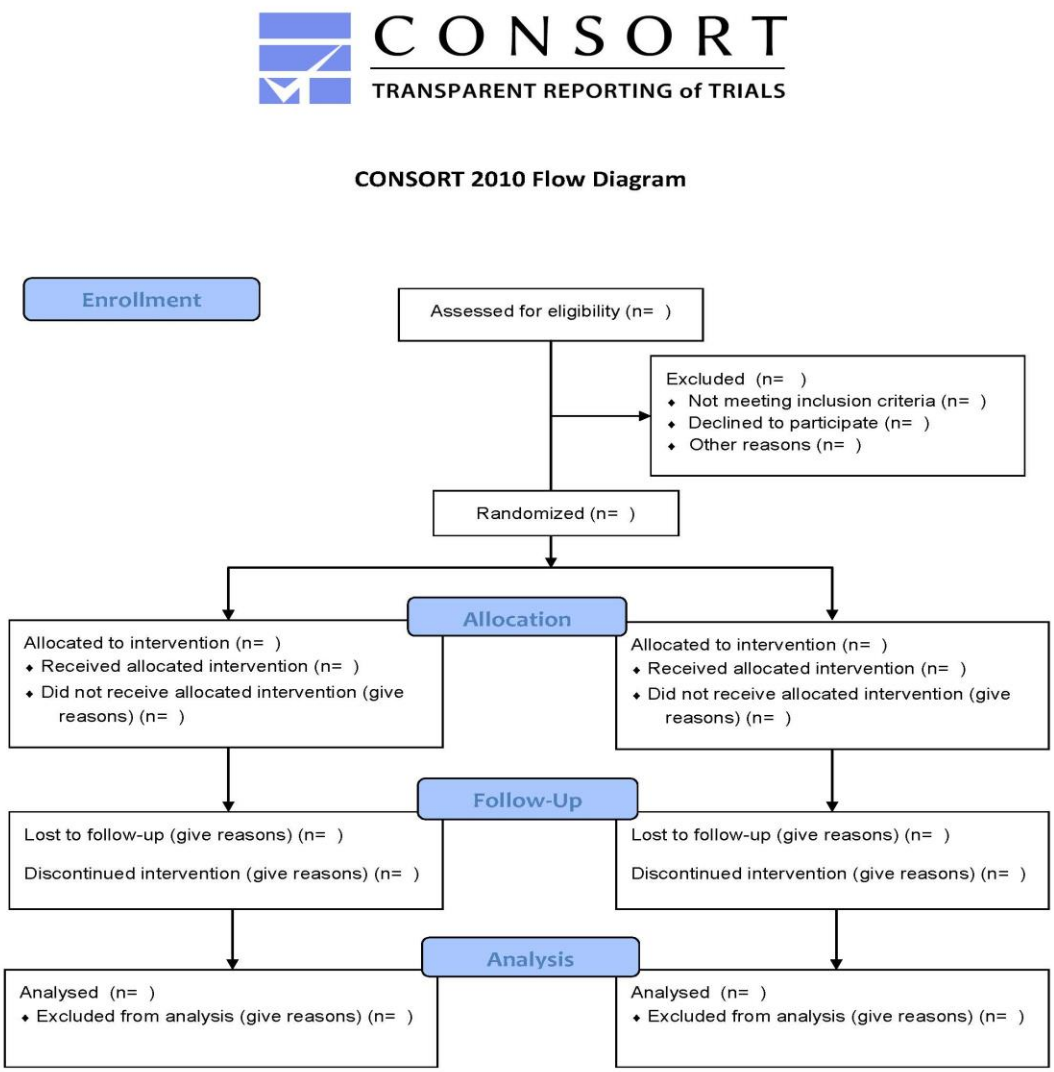


Study design. The trial is a multicenter randomized pragmatic clinical trial comparing the safety of early versus late IH repair in premature infants who are diagnosed with an IH while in the NICU, or other nursery, at a participating site.[89] Randomization will be stratified according to site and GA strata (<28 weeks and ≥28 weeks estimated GA) to further ensure as equal risk groups as possible.

Patient recruitment and educational component. Based on prior experience with the Necrotizing Enterocolitis Surgery Trial (NEST), we understand that randomization of surgical therapies is difficult for parents and providers. SUBJECT LOCATOR will be used as a recruitment tool to identify potential research subjects based on data available in VU clinical systems (e.g. STAR Panel, WizOrder, Clinic Scheduling). The SUBJECT LOCATOR program is part of a toolset available through VICTR that enables teams to specify inclusion/exclusion criteria for a specific study. The inclusion/exclusion criteria are codified for computable use and combined with data coming through VU Clinical Systems to proactively identify individuals who might qualify for a study. Once a 'match' is made, research study personnel are alerted using confidential messaging or a secure web portal and they may then use the information to further review the patient's information using the Vanderbilt Electronic Medical Record. If the patient is further deemed a candidate for the study, the study personnel will notify the patient's care provider who will then ask the patient if they would be interested in communicating with study personnel about research population. We will encourage at least one educational visit to the parents of each eligible infant to discuss the risks and benefits associated with early and late IH repair, the importance of the trial, and the requirements of participation. After parents have received baseline information about the risks associated with an IH and its treatment, they will be better informed prior to deciding about participation in the HIP trial. The primary goal of this strategy is to promote patient recruitment, adherence to the protocol, and full understanding of the rationale for the trial. This has been used successfully in prior pediatric trials.[90] We expect >60% of eligible patients to be successfully randomized, based on the pilot trial and another similar trial conducted by Dr. Blakely.[14,42] Participating sites were carefully screened and selected to ensure surgeon willingness to randomize patients. Prior performance of trials and willingness to randomize infants in the NEST trial were important considerations. We are using conservative estimates of available patient numbers to estimate the 2.5-year enrollment period, since there are likely >500 infants eligible each year across all sites, as based on two data sources. The enrollment period could well be reduced if these numbers are stable and we achieve a reasonably high randomization rate, as expected.

Every attempt will be made to follow the guidelines put forth by the Consolidated Standards of Reporting Trials (CONSORT) Statement. [105] These guidelines, which are comprised of a 25-item checklist and a flow diagram, are considered the gold standard to which all randomized clinical trials are held and required by most peer-reviewed journals when publishing findings of RCTs. The expected progress of participants is visually presented in the flow diagram (Figure 2).

Figure 2. CONSORT 2010 Flow Diagram



CONSORT ensures proper and consistent reporting of RCTs and that the sample enrolled is an accurate representation of the population as a whole. To better understand the patient population, anonymous, de-identified information will be housed in the REDCap database and will include:

- The number of patients screened for the study
- The gestational age and gender of screened infants
- The number of eligible patients
- For those ineligible, the reason(s) why
- For those eligible but not enrolled, the reason(s) why and the planned treatment (early or late IH repair)

This de-identified, anonymous data will demonstrate whether or not the randomized cohort is similar or different to the overall pool of participants. Thus, we will be able to determine whether or not the resulting findings of the RCT are generalizable. The coordinator and PI at each site will communicate regularly with their institution's surgeons, neonatologists, research support personnel, residents, fellows, nurses and surgery schedulers to ensure they are aware of babies that present with an inguinal hernia so that all applicable infants are screened for possible participation.

Enrollment and Randomization. Every effort will be made to obtain signed informed consent by the parent(s)/guardian(s) as soon as possible. When the surgeon or neonatologist determines the baby may be a good candidate for the HIP trial but parent/guardian is unavailable for written consent, phone consent will be obtained. The research coordinator or PI will provide the parent with the study information and answer any questions the parents have. If parental consent is given over the phone, the baby will be randomized to treatment group so that providers can determine the treatment plan for the participant. The consent process will continue when the parent is again at the hospital where written consent will be obtained. Written consent will be obtained no later than the day of surgery for the IH repair. After obtaining informed consent, research coordinators will create a new patient record in the REDCap database, enter patient data, and engage the REDCap randomization module. This easy to use interface will provide an immediate treatment group assignment, which will be stratified according to site and GA strata. Enrollment will occur when the infant is approximately 2 weeks from anticipated NICU discharge. This is the exact time at which the decision is currently made (outside any study) whether to perform the IH repair or defer until later.

HIP Trial Setting. Participating clinical sites have been carefully selected based on a high number of premature neonates with an IH, equipoise of surgeons regarding the issue being studied, and the necessary infrastructure to complete the trial (surgeon and neonatology site PI and research coordinator and/or fellow).

Table 2. Clinical Participating Sites and Site Surgery Investigators

*Eligible patient population calculated from site-specific data and data for 2011 from Pediatric Health Information System (PHIS) (available for 13/19 [68%] sites).

Bold = sites that currently have resources for neurodevelopmental follow-up as part of Neonatal Research Network

Hospital	University Affiliation	Site PI
Akron Children's Hospital	n/a	Avaraham Schlager, MD
Alabama Children's Hospital	University of Alabama Birmingham	Robert Russell, MD, MPH
Albany Medical Center	Albany Medical College	Elizabeth Renaud, MD
All Children's Hospital	Johns Hopkins Medicine	Nicole Chandler, MD
Amplatz Children's Hospital	University of Minnesota	Brad Segura, MD, PhD
Arkansas Children's Hospital	University of Arkansas	Sid Dassinger, MD
Naval Medical Center San Diego	Uniformed Services University of the Health Sciences	Donald Lucas, MD, MPH
Children's Hospital at Erlanger	University of Tennessee Health and Science Center at	Curt Koontz, MD

	Chattanooga	
Children's Hospital of Colorado	University of Colorado Denver	Timothy Crombleholme, MD
Children's Hospital of Los Angeles	University of Southern California	David Bliss, MD
Children's Hospital of Michigan	Wayne State University	Justin Klein, MD
Children's Hospital of Wisconsin	Medical College of Wisconsin Milwaukee	Casey Calkins, MD
Children's Medical Center Dallas	University of Texas Southwestern	Adam Alder, MD
Children's Memorial Hermann Hospital	University of Texas at Houston	Kuojen Tsao, MD
Children's Mercy Hospital	University of Missouri – Kansas City	Shawn St. Peter, MD
Children's National Medical Center	n/a	Andrea Badillo, MD
Cincinnati Children's Hospital Medical Center	University of Cincinnati	Roshni Dasgupta, MD
Columbia University Medical Center	Columbia University	Steven Stylianos, MD
Connecticut Children's Medical Center	University of Connecticut School of Medicine	Richard Weiss, MD
Dartmouth-Hitchcock Medical Center	Geisel School of Medicine	Daniel Coitoru, MD
Dayton Children's Hospital	n/a	Arturo Aranda, MD
Duke Children's Hospital	Duke University	Henry Rice, MD
East Tennessee Children's Hospital	University of Tennessee Knoxville	Erik Jensen, MD
Floating Hospital for Children	Tufts University	Jonathan Davis, MD
Golisano Children's Hospital	University of Rochester	Walter Pegoli, MD; Carl D'Angio, MD
Harbor Medical Center	University of California Los Angeles	Daniel DeUgarte, MD
Hurley Medical Center	n/a	Masih Kader, MD
LeBonheur Children's Hospital	University of Tennessee Health and Science Center Memphis	Eunice Huang, MD
Mattel Children's Hospital	University of California Los Angeles	Daniel DeUgarte, MD
MUSC Children's Hospital	Medical University of South Carolina	Aaron Lesher, MD
Nationwide Children's Hospital	Ohio State University	Katherine Deans, MD
OHSU Hospital	Oregon Health and Science University	Kenneth Azarow, MD
Primary Children's Hospital	University of Utah	Steven Fenton, MD

Rady Children's Hospital	University of California San Diego	Karen Kling, MD
Rainbow Babies & Children's Hospital	Case Western Reserve University	Edward Barksdale, MD
Rocky Mountain Hospital for Children	n/a	Bethany Slater, MD
Seattle Children's Hospital	University of Washington	Adam Goldin, MD, MPH
Southern California Permanente Medical Group, Kaiser	n/a	Donald Shaul, MD
St. Louis Children's Hospital	Washington University	Jacqueline Saito, MD, MSCI
Texas Children's Hospital	Baylor College of Medicine	Monica Lopez, MD
University of Iowa Children's Hospital	University of Iowa	Joel Shilyansky, MD
UVA Children's Hospital	University of Virginia	Jeffrey Gander, MD
Women & Children's Hospital of Buffalo	University at Buffalo	David Rothstein, MD
Women and Infants Hospital of Rhode Island	Brown University	Arlet Kurkchubasche, MD

Study population. During the 2.5-year enrollment period, 615 patients will be randomized to either early or late IH repair, after obtaining informed consent from the parent or legal guardian. Data from all eligible patients will be entered into the REDCap database as well as their enrollment status. For infants that were not randomized, the reasons for this will be recorded (e.g. parent or provider refusal, or other reasons).

Inclusion Criteria: 1) GA < 37 wks, 0 days; 2) In a NICU / other nursery, at participating sites; 3) Diagnosed with an IH per the pediatric surgeon; and 4) Parental consent and providers willing to randomize the infant.

Exclusion Criteria:
1. Associated factor that impacts timing of repair (e.g. clinical factors that might preclude early IH repair) 2) Infant is undergoing another operative procedure and IH repair is planned as a secondary procedure such that timing of IH repair is impacted (e.g. fundoplication or gastrostomy tube is planned, and IH repair is considered a secondary procedure); 3) Known major congenital anomaly or chromosomal abnormality; and 4) Family unable to return for follow up and late IH repair; or likely unable to monitor IH as outpatient (e.g. maternal incarceration or moving away from the area).

Study intervention.
Treatment Groups.
a. **Early IH Repair** = IH repair is performed promptly after randomization and prior to NICU discharge; ~38 weeks PMA, based on pilot trial and experience in practice.

b. **Late IH Repair** = IH repair will be performed at a goal of 5 months after NICU discharge (~ 55 - 60 weeks PMA). This PMA was selected as the goal time for late repair because this reflects current surgical care and most Children's Hospitals *consider* allowing the operation to be done as an outpatient at this time. The decision regarding admission will be determined by the providers for each infant according to site guidelines.

A small percentage of infants in either treatment group will likely undergo IH repair at times other than the goal times, due to clinical considerations. For example, some infants assigned to late IH repair will undergo repair prior to 55-60 weeks PMA due to hernia incarceration. These occurrences are important outcome measures that will be recorded. It is also possible that some infants assigned to early IH repair will undergo IH repair after discharge due to changes in the infant's condition after randomization.

Study Procedures and Outcome Determination.
After treatment group assignment, each infant will receive care according to the practice at each site, in keeping with a pragmatic trial design. Existing treatment practices that will be allowed include different operative approaches (laparoscopic or open IH repair), anesthesia management (general or regional or other), and other treatment considerations. All details will be documented for analysis. There are no prior studies indicating an impact on the HIP trial outcome measures based on these differences in treatment.

Data will be collected from these broad temporal categories: demographics, perinatal, NICU course prior to enrollment, surgical consult just prior to enrollment, status of infant at enrollment, information at NICU discharge, intraoperative details (includes surgery and anesthesia-specific data), outpatient data and data at the conclusion of the first phase of the trial (10 months after randomization). Ten months was selected as the end of SAE and hospital days determination by assuming that the earliest PMA at NICU discharge would be at 35 weeks. An infant discharged at 35 weeks PMA who had very late IH repair at 65 weeks PMA would still have a 1.5 - 2 month post-operative observation period. The aim is to have the same observation period for infants in both treatment groups to decrease ascertainment bias. Infants in the early IH repair group will have a longer observation period postoperatively, whereas those in the late IH repair group have a longer observation period preoperatively. This study design represents clinical practice and will allow the detection of SAE related to performing or delaying IH repair, which is our goal. After the first phase of the study (10 months after enrollment), surveillance for SAEs will cease and the patient will then enter the "tracking phase" in preparation for neurodevelopmental assessment at 22-26 months corrected age (chronological age - # weeks / months premature).

Primary / secondary outcomes definitions.
Ascertainment of Primary Outcome (proportion of infants with >1 SAE in the 10 months after enrollment).
Specific definitions (based on standard definitions when available) are listed in the Appendix for all measured SAEs. Trained coordinators will prospectively monitor all trial patients for SAEs from enrollment until 10 months afterwards. Measures to avoid bias and maximize reliability in assessing SAEs will include field testing of these definitions, discussing definitions in conference calls with the coordinators to identify and resolve problems in their assessment across different sites and settings, and assessing inter-coordinator reliability for a random 10%. From the start of the IH repair operation until 24 hours post-operatively, the coordinators will ascertain whether an SAE has occurred by briefly asking the clinical nurse, anesthesia provider, and surgeon if they are aware of any SAE, in addition to medical record review. This face-to-face, very brief interaction (~5 minutes) has been shown to reliably track SAEs in prior studies.[91] A primary outcome adjudication committee, blinded to treatment group assignment, will review all case report forms during the trial and will make the final assessment of SAE occurrence. Additionally, the committee will assign the highest level of harm for patients that have had $\geq$1 SAE according to the National Coordinating Council for Medication Error Reporting and Prevention (NCC MERP), as a secondary outcome.[92-94] Categories E-I, in this Index will be utilized, as these categories describe levels of adverse events associated with patient harm (See Appendix for Index). The NCC MERP index is primarily focused on medication *errors*, rather than *any* SAE as we are measuring. However, categorization of SAEs according to the level of patient harm, if any, is likely helpful in fully understanding the morbidity in HIP trial infants.

Cardiorespiratory or Anesthesia – related SAE	Inguinal Hernia or Surgery – related SAE
Death	IH incarceration
Use of extracorporeal membrane oxygenation (ECMO)	IH strangulation
Cardiac arrest	IH recurrence
Cardiopulmonary resuscitation	Reoperation for IH or related to initial operation
Hypotension treated with vasoactive drugs	Injury to adjacent structure / organ
Apnea treated with increased respiratory support	Wound disruption (deep)
Local anesthetic toxicity (e.g. seizure)	Surgical site infection (deep)
Prolonged intubation / ventilation (>48 hours post op)	Other SAE involving intervention to prevent or address patient harm
Unplanned reintubation	

Intraoperative unplanned extubation	

Table 3. Serious Adverse Events To Be Identified and Tabulated In HIP Trial.

<u>Neurodevelopmental Follow-up Assessment.</u>
The NICHD U01 award provides limited funding for the first 200 participants with access to robust neurodevelopment follow-up programs. To increase validity and efficiency and reduce costs, we will conduct the follow up assessment in the 8 study centers in the NICHD Neonatal Research Network (**NRN**) that already have certified examiners, high follow-up rates, and well demonstrated success in assessing developmental effects in clinical trials (UT Houston, UT Dallas, UAB, UCLA, Children's Mercy Hospital, Nationwide Children's, Duke, U Michigan). These selected sites have protocols for annual training of examiners to ensure reliability of assessments.[95] In addition to the 8 NRN sites, the neurodevelopmental assessment will also be performed at other participating sites with established neurodevelopmental programs that are capable of administering a neurologic exam and the Bayley Scales of Infant Development, 3rd Edition (BSID III) with current resources. Participants will complete the follow-up assessment at 22-26 months correct age. The NICHD U01 award provides limited funding for the first 200 participants. Additional funding is being sought to promote neurodevelopmental follow-up assessment of all HIP infants. HIP sites outside of the NRN were selected partly based on the presence of a neurodevelopmental PI at the site with the intention of implementing neurodevelopmental follow-up once funding is secured for all participants. All participants volunteering for the neurodevelopmental portion of the study will be seen by certified examiners at their site's neurodevelopmental follow-up clinic. If a HIP trial infant isn't eligible for 2 year neurodevelopmental follow-up (e.g., very late preterm infant [36 weeks gestation] with no comorbidities) and the family does not wish to return for follow-up, this preference will be respected and no in-person follow-up obtained. In this case, if parents are willing and should funding allow, a brief phone conversation will occur where trained personnel will ask standard questions as part of routine neurodevelopmental follow-up.

The Bayley Scales of Infant Development, 3rd edition, (BSIDIII) will be administered[96] and a neurologic exam will also be performed. All assessments will be performed by NICHD NRN certified examiners or examiners certified by their respective site. The BSID III Cognitive, Language and Motor scales will be administered as part of the routine NRN 22-26 month corrected age visit. The BSIDIII offered at the other participating sites are available to all patients within their infant follow-up clinics. The cognitive component of the BSIDIII is hypothesized to differ according to treatment group assignment in the HIP trial and is the basis for the sample size calculation below. The Composite scores for each of these scales are age-standardized with a mean score of 100 and standard deviation of 15; a score of less than 70 (>2 SDs below the mean) indicates significant delay. Children with severe neurologic and/or developmental delay as to preclude Cognitive scale administration will be assigned a Composite score of 54. Cerebral palsy will be defined as a non-progressive disorder characterized by abnormal tone in at least 1 extremity and abnormal control of movement and posture[97], and the Gross Motor Function Classification System will be utilized to classify severity.[98] Data from these assessments will be entered into the REDCap database by existing follow-up program coordinators at each site.

At the time of the 22-26 month in-person follow up visit or phone follow-up, the child's parent or primary caregiver will be asked about other operations that may have occurred in the interval between the end of primary outcome ascertainment (10 months after enrollment) and the follow-up assessment.

Statistical Analysis Plan (Claudia Pedroza, PhD, primary study statistician).
<u>Sample size</u>: To test the primary hypothesis that overall SAE rates will be 30% in the early repair group compared to 20% in the late repair group 293 infants per group (N = 586 total) will be required to have 80% power (two-sided alpha of 0.05) to detect a 10% absolute difference. We conservatively estimate 20 eligible infants/year at each site with a 60% randomization rate and 95% follow-up rate through 10 months post randomization. With 19 participating sites we estimate randomizing 615 infants (with 586 assessed) in ~ 2.5 years. For the secondary outcomes, we will have 98% power to detect a median difference of 3 hospital days (assuming median=8, mean=18 for early group and median=5, mean=13 for the late group and a log normal distribution). For BSIDIII scores, we will be able to detect a 0.4 SD difference (6 points on the BSIDIII)

assuming that 200 infants are assessed at 22-26 months. Table 4 shows power according to different sample sizes. Power will be increased as more infants complete the neurodevelopmental follow-up portion of the study.

Table 4. Power to detect hypothesized effect sizes under varying number of infants.

	Power	
Infants / group	10% Difference in SAE rates	3 Day Difference in Median Hospital Days
293	80%	98%
250	73%	97%
200	64%	92%
150	52%	81%

The data will be analyzed using intent-to-treat analyses. Primary and secondary outcomes will be analyzed using general linear mixed models (GLMMs) with both continuous and categorical outcomes as dependent variables. We will examine all variables to determine that they are relatively symmetric and unimodal, examine group variances for obvious deviations, and examine residuals from each model to ensure reasonable adherence to the assumptions of the models. We will evaluate the comparability of our groups on key demographic variables and will adjust for them if necessary (e.g., gender, ethnicity).

For the primary outcome of the % of infants with $\geq$1 SAE, we will use a GLMM (binomial with log link) with treatment group (early, late) and GA (<28; $\geq$28 wks) stratum as covariates and center as a random effect. Hospital days will be analyzed with a lognormal model using treatment group and GA as covariates (center as a random effect). A linear mixed model will be used for BSIDIII scores with same covariates and random center. The absolute risk difference (which is better understood than relative risk by clinicians and patients) and 95% confidence interval (CI) will be reported for the primary outcome. For continuous outcomes, we will report absolute differences and 95% CIs. Similar models will be used to analyze all other secondary outcomes.

In addition to the planned final frequentist analysis, we will also conduct a final Bayesian analysis of the primary outcome and major secondary outcomes. All analyses will be conducted using neutral priors centered at relative risk of 1.0 with 95% credible interval of 0.33-3.0.

Treatment Effect Modifiers. To investigate treatment effect modifiers for the primary and secondary outcomes (hospital days and BSIDIII scores), we will use Bayesian hierarchical models with terms of interaction between treatment group (early, late) and the prespecified potential moderators: GA (<28; $\geq$28 wks) bronchopulmonary dysplasia (yes/no), maternal education $\leq$12 yrs (yes/no), prenatal care (yes/no), IH difficult to reduce (yes/no). The conservative Bayesian approach of Dixon and Simon[99] allows us to specify a priori how likely (or unlikely) it is for subgroup differences to be present and to shrink the subgroup estimates to the overall mean treatment effect. Prior distributions for main effects will be neutral ~N(0,1000) and Uniform(0,1000) for SD parameters. Neutral and skeptical priors will be used for interaction terms. Point estimates of treatment effect and 95% credible intervals will be reported along with probability of treatment benefit/harm. We will conduct similar subgroup analyses to examine differences in treatment effect according to gender, race/ethnicity, although such differences are not expected (required for NIH Phase III RCTs).

Construction and Testing of a Prediction Tool. We will use the final subgroup Bayesian models to develop and test a potential web-based tool. The models will include only covariates defining subgroups that have a differential treatment effect (i.e., large probability of interaction between covariate and treatment group). This tool will be designed to help clinicians and parents assess the risks and benefits of early and late repair with regards to SAEs, hospital days, and Bayley scores < 70 (impairment) based on the patient characteristics noted above. Validity will be rigorously assessed in the study population using methods similar to that previously used by Tyson et al (to help assess the effects of intensive and comfort care on various outcomes for extremely premature infants).[83] We will use posterior predictive checks (i.e., simulated data sets from the final model are compared to observed data for major discrepancies) and ROC curves (where appropriate) to assess the accuracy of the prediction tool relative to that of assuming the overall better timing of IH repair is better for all patients.[100] To guide decision-making, the web-based tool will take patients' characteristics and provide estimated probabilities of SAEs, number of hospital days, and probability of impairment with early and late repair. To augment parental comprehension, probabilities will be represented graphically as well as with percentages.[101] These analyses are considered novel and secondary to the main purpose of the HIP trial. To the extent that significant differences are found related to specific patient subgroups, these analyses may help implement evidence-based surgical decision making.

Interim Analyses and Early Stopping Rules: The primary statistician (Dr. Pedroza) will conduct a safety and efficacy analysis in December 2019, when we expect to have 10-month outcome data for approximately 200 infants. Bayesian methods will be used to calculate the probability of harm using the pre specified outcomes of death and prolonged hospital stay >30 days post surgery. We will also conduct an efficacy Bayesian interim analysis to calculate the probability of benefit on the primary outcome, proportion of infants with an SAE, and secondary outcome of total hospital length of stay. We will use neutral priors that a priori assume a 50-50 likelihood of either repair management being better. In contrast to frequentist analyses, Bayesian analyses provide direct probabilities of benefit and may be more interpretable by clinicians. In deciding whether to stop or continue the trial, we define the following stopping guidelines:

a. Stopping for safety: probability of increased harm >90% for either arm (for either death or prolonged hospital stay)
b. Stopping for efficacy: probability of decreased SAEs >95% for either arm

These analyses and data reports will be presented to the DSMB, which will monitor the study for major differences in safety between the treatment arms.

Compliance and retention strategies. We will utilize other evidence-based strategies to increase adherence to the trial protocol.[102] A risk with the HIP trial is parental desire to request IH repair according to their preferences rather than according to the study protocol. This is most likely to occur in infants assigned to late repair (i.e. parents may want repair prior to 55-60 weeks PMA for a variety of reasons). Using clear and simple written descriptions, we will develop a Parent Handbook with information about the HIP trial, including a list of names and office numbers of the research team who may be contacted with questions and in case of emergencies. The handbook will include important dates that will be personalized for each participant (written in at the site level) regarding follow up appointments. We will include information regarding the importance of keeping the research team updated with contact number changes. Sites will be encouraged to create materials that will assist them in educating families involved in the HIP trial, e.g. parent handbook. The key to the success of retention and follow-up will be compliance, attention to detail, consistency of the research team and trust among the parents and research teams.

Every attempt will be made by the HIP Trial Study team to contact participating families regularly and keep them from becoming lost to follow up. Identifying a stable contact person (grandparent, neighbor, friend) not living with the enrolled infant but who will always know the family's location will assist in tracking patients and decrease the number lost to follow up. We will also encourage these individuals to participate in the parent education visit prior to randomization in the trial. Drs. Tyson and Hintz have extensive experience in long-term outcomes studies, including devising and implementing tracking strategies and assuring excellent follow-up rates, and they will contribute to creating tracking and follow-up mechanisms.

We will also have a dedicated central phone number and email address that will be accessible 24-hours/7 days a week for urgent questions or concerns from parents, providers, or anyone associated with the HIP trial which will be answered by either Dr. Blakely (PI) or Andrea Krzyzaniak, MA (lead study coordinator). Having an accessible source of information will promote proper determination of eligibility of infants, proper randomization, and adherence to the study protocol, as well as decrease loss to follow up.

The HIP trial website will serve as a tool for collaborators, participants, and potential participants to find general information about the study and contact information. Website hosting will be with Pressable (www.pressable.com) and the URL will be www.HIPTrial.org. The website will be user-friendly and provide information about the trial in lay terms. A brief introduction to the study and study contact information will be available on the Home Page. Additional tabs will provide information on the study investigators, participating sites, and associated contact information; a map illustrating where participating centers are located; resources such as fact sheets on inguinal hernias and premature infants; and links to other reputable online resources such as ClinicalTrials.gov and NIH.gov as well as links to participating site's websites. A password-protected portion of the website will be available for collaborators at participating sites and accessible only to users with a login. This portion of the website will house essential study forms including the approved protocol, consent forms, manual of operations, and a link to the REDCap website where collaborators will need to login to access the study's database. Information on the website will be kept up-to-date by the research coordinator. No PHI will be placed on the website at any time.

Strategy if sufficient patient accrual goal is not met. As detailed elsewhere, we are conservatively expecting 240 randomized patients / year in years one and two, and 120 patients in the first half of year three to reach the required sample size on time. There is a reasonable likelihood that we will meet this number sooner, as more surgery consults are generated for eligible infants during the NICU period, rather than discharge and referral back to the surgeon without a consult. This change in practice has already occurred at several sites during the design phase of the HIP trial. If the numbers of eligible infants increases over the numbers provided prior to HIP trial design, enrollment will also increase. We have clear recruitment goals and the actual versus expected enrollment will be on the agenda for every monthly conference call between HIP trial investigators and this information will also be distributed separately by the HIP coordinating center. In the event that participating centers do not accomplish the enrollment goals, every effort will be made to examine the reasons for insufficient recruitment and adjustments will be made to ensure that the overall study enrollment is achieved. If a given site randomizes below 40% of eligible patients at the six-month period of the trial, additional calls will be scheduled to review site-specific practices and site visits will be conducted. If poor randomization continues, additional replacement sites will be considered. We do not expect enrollment to take more than 2.5 years, but we are prepared to continue enrollment until the sample size is met (at the same budgeted cost). Our proposed budget is meant to incentivize rapid enrollment, as the financial support to surgeon site PIs is only via capitation for randomized infants with full data entry into the REDCap database.

Interpretation of Results in Formulating Conclusions. Large RCTs are needed to provide the most rigorous and unbiased safety assessments for major interventions, including IH repair as the most common major surgical procedure for premature infants. The proposed trial will be subsidized by the participating institutions, led by investigators exceptionally experienced in successfully designing and conducting difficult RCTs at relatively low cost, and is designed to be the largest RCT feasible under an R01 mechanism. We expect adequate power to assess differences between groups in SAEs and high power to assess differences in hospital days. While our sample size estimates reflect our honest expectations based on multiple data sources, we acknowledge that enrollment rates might not met expectations. If enrollment is less than expected, we could stop the study early, resort to observational studies highly susceptible to bias, or choose to do no study at all, a reason that the hazards of widely use therapies may go unsuspected for years.[8] As Schulz and Grimes note, unbiased trials with imprecise results trump no results at all.[103] We propose instead to use frequentist and Bayesian analyses as described above and base our conclusions about safety on the totality of evidence from the study. A significant overall benefit for a major secondary outcome (e.g. number of hospital days during the study period) can be a legitimate basis for recommending a therapy providing the primary outcome (e.g. proportion of infants with $\geq$ 1 SAE) and any other important outcome is in the same direction. For this reason, we would recommend that late IH repair be routinely performed if it was associated with a significant reduction in total hospital days as long as the SAEs and Bayley III scores did not favor the early group, particularly if there were a >80% posterior probability that SAEs were less common and Bayley III scores higher with late than early IH repair.

Cessation of Trial: After completing the trial and conducting the analyses, the parents of enrolled infants, as well as participating providers, will be informed of the results of the trial (an executive summary of the results) within a timely manner (<8 weeks after such data are available).

HIP Trial Organization and Governance. The HIP trial will have 8 organizational components: (1) 20-30 participating clinical sites across the United States; (2) the Clinical Coordinating Center; (3) the Data Coordinating Center (DCC); (4) the Trial Principal Statistician; (5) Executive Committee; (6) Steering Committee, consisting of Site Investigators and Publication Subcommittees; (7) External Advisory Committee; and (8) Data Safety and Monitoring Committee. The Chairs of the Clinical Coordinating Center and the DCC are located at Vanderbilt University Medical Center in Nashville, TN. The principal statistician is located at the University of Texas Houston Health Science Center in Houston, TX.

Table 5. HIP Trial Investigator Team

Investigator	Area of Expertise	Role on Project
Martin Blakely, MD, MS*	Pediatric Surgery	PI
Jon Tyson, MD, MPH*	Neonatology, Epidemiology	Co-I
Susan Hintz, MD, MS*	Neurodevelopmental Follow Up	Co-I
Claudia Pedroza, PhD*	Bayesian and frequentist statistics	Co-I
Eric Thomas, MD, MPH	Patient Safety Research	Co-I
Stephany Duda, PhD	Director, Data Coordinating Center	Co-I
John Brock, III, MD	Pediatric Urology	Co-I
Stephen Hayes, MD, Andreas Loepke, MD	Pediatric Anesthesia	Co-I
Nathalie Maitre, MD, PhD	Neonatology	Co-I
Katherine Deans, MD	Pediatric Surgery	Co-I
Michael DeBaun, MD, MPH	RCT Design / Implementation	Co-I
Lillian Kao, MD, MS	Surgical trial methodology	Co-I

*Executive committee

Data Coordinating Center (DCC): Dr. Stephany Duda will direct Vanderbilt Bioinformatics (DCC) along with DCC program director, Lauren Davis, and a project team of PhD and Master's level statisticians, database programmers, and other personnel. Drs. Blakely and Duda will continue to work together to ensure the highest quality data collection and consistency between sites. The DCC will provide monthly enrollment reports to all sites, which will be facilitated by a HIP Trial website, developed with support from the DCC). Sites will inform the DCC of user changes and the DCC will make associated updates to REDCap. This will help ensure only approved study personnel have access to the database. Additionally, the HIP study database will have quality checks performed every 30 days. HIP coordinating center personnel will review study data and send queries to sites via REDCap when identifying discrepancies. Resolution of any discrepancies will be expected within 30 days.

Data Safety and Monitoring Committee (DSMC). Michael O'Shea, MD, MPH (Chief, Neonatal-Perinatal Medicine at University of North Carolina Children's; Distinguished Professor of Pediatrics) – chairman. Jeffrey H. Haynes, MD (Professor of Pediatrics; Pediatric Surgeon; Director Children's Trauma Center, Children's Hospital of Richmond at Virginia Commonwealth University) – member. Dennis Wallace, PhD (statistician, RTI International)- member. The committee will meet twice/year throughout the study period and review data provided by the DCC.

Dissemination / Implementation. Dr. Blakely has completed a manuscript invited by the Archives of Diseases in Childhood addressing IH repair in premature infants and is preparing) a methods paper addressing the challenges / opportunities in designing a large multicenter surgical safety trial that includes innovative strategies: use of the REDCap database / randomization module, measures to augment SAE reliability and validity, and Bayesian analyses of effect modifiers. The primary manuscript will be submitted by the end of year 3 and the manuscript describing neurodevelopmental findings by the end of year 5 (both to the N Engl J Med or JAMA). Secondary manuscripts will address process measures, center differences, development of risk estimator, and likely others. Trial results will also be presented at national surgery, neonatology, anesthesia, and quality improvement/safety meetings. We will seek novel dissemination methods, e.g. webinars and e-mail distributions (for example, the monthly "e blast" from the American Pediatric Surgery Association to its members). If, as we expect, we successfully validate our tool to estimate and compare the risks of early and late IH repair, we will make it available on the internet to promote better informed decisions by professionals and parents, as Dr. Tyson has successfully done to better inform decision-making for extremely premature infants.

Institutional Board Review: The trial will be submitted for full IRB review at each participating institution. The trial has been granted full IRB approval at Vanderbilt (IRB #121573).

Trial Commencement: The HIP trial was started as a pilot phase at Vanderbilt only June 3, 2013. The purpose of this phase is to test the REDCap data randomization module, data capture methods, establish the working relationship of Dr. Blakely with the DCC, as well as gain important experience in approaching families of eligible infants. After starting the pilot phase at Vanderbilt, 5 other sites have also initiated the trial at their sites as a part of this pilot phase (University of Texas at Houston, University of Tennessee at Memphis, Duke University, Washington University, and the University of Michigan). During the pilot phase, data for non-randomized, but eligible infants was not sought.

Bibliography

1. Leape LL, Berwick DM. Five years after To Err is Human: what have we learned? JAMA 2005;293:2384-2390.

2. Institute of Medicine (Ed.). (1999). To err is human: building a safer health system. Washington, D.C.: National Academy Press. http://www.iom.edu/reports/1999/To-Err-is-Human-Building-A-Safer-Health-System.aspx

3. Moss RL, Henry Mc, Dimmitt RA, Rangel S, Geraghty N, Skarsgard ED. The role of prospective randomized clinical trials in pediatric surgery: state of the art? J Pediatr Surg 2001;36:1182-1186.

4. Blakely ML, Kao LS, Seetharamaiah R, Tsao K, Huang EY, Lally KP. A critical appraisal of published randomized trials within pediatric general surgery during the time period 2000-2009. American Pediatric Surgical Association, annual meeting, May 2011.

5. Lacchetti C, Ionnidis J, Guyatt G. Surprising results of randomized trials. *in* Users' Guide to the Medical Literature. A Manual for Evidence Based Clinical Practice., 2nd edition. AMA Press, 2008, pp.113-40.

6. Prasad V, Cifu A, Ioannidis JP. Reversals of established medical practices: evidence to abandon ship. JAMA 2012;307:37-8

7. Chalmers I, Sinclair JC. Promoting perinatal health: is it time for a change of emphasis in research? Early Hum Dev. 1985;10:171-91.

8. Silverman WA. Human experimentation in perinatology. Clin Perinatol. 1987;14:403-16.

9. Silverman WA. Medicine's "therapeutic imperative"- haunted by the potential for tragedy. Paediatr Perinat Epidemiol 2004;18:318-20.

10. Kennedy KA, Tyson JE (eds). Evidence-Based Medicine. Clin Perinatol 2003;30:xv-xvi.

11. Sekar KC. Iatrogenic complications in the neonatal intensive care unit. J Perinatol. 2010;30 Suppl:S51-6.

12. Peevy KJ, Speed FA, Hoff CJ. Epidemiology of Inguinal Hernia in Preterm Neonates. Pediatrics 1986;77:246-247.

13. Baird R, Gholoum S, Laberge JM, Puligandla P. Prematurity, not age at operation or incarceration, impacts complication rates of inguinal hernia repair. J Pediatr Surg 2011;46:908-11.

14. Blakely ML, Neville HL, Mehall JR, Smith SS, Warner BW, Mantor PC, Cox CS, Jr., Lally KP. A comparison of early versus delayed inguinal hernia repair in premature infants. Presented at Pacific Association of Pediatric Surgeons, La Jolla, CA, May 2002.

15. Zamakhshary M, To T, Guan J, Langer JC. Risk of incarceration of inguinal hernia among infants and young children awaiting elective surgery. CMAJ 2008;179:1001-1005.

16. Vaos G, Gardikis S, Kambouri K, Sigalas I, Kourakis G, Petoussis G. Optimal timing for repair of an inguinal hernia in premature infants. Pediatr Surg Int 2010;26:379-385.

17. Lautz TB, Raval MV, Reynolds M. Does timing matter? A national perspective on the risk of incarceration in premature neonates with inguinal hernia. J Pediatr 2011;158:573-577.

18. Wiener ES, Touloukian RJ, Ridgers BM, Grosfeld JL, Smith EI, Ziegler MM, Coran AG. Hernia survey of the section on surgery of the American academy of pediatrics. J Pediatr Surg 1996;31:1166-1169.

19. Antonoff MB, Kreykes NS, Saltzman DA, Acton RD. American Academy of Pediatrics section on surgery hernia survey revisited. J Pediatr Surg 2005;40(6):1009-1014.

20. Wang KS and American Academy of Pediatrics Committee on Fetus and Newborn and Section on Surgery. Assessment and Management of Inguinal Hernia in Infants. Assessment and management of inguinal hernia in infants. Pediatrics 2012;130(4):768-773.

21. Uemura S, Woodward AA, Amerena R, Drew J. Early repair of inguinal hernia in premature babies. Pediatr Surg Int 1999;15:36-39.

22. Allen GS, Cox CS, White N, Khalil S, Rabb M, Lally KP. Postoperative respiratory complicaions in ex-premature infants after inguinal herniorrhaphy. J Pediatr Surg 1998;33:1095-1098.

23. Lau ST, Lee YH, Caty MG. Current management of hernias and hydroceles. Seminars in Pediatric Surgery 2007;16:50-57.

24. Gollin G, Bell C, Dubose R, Touloukian RJ, Seashore JH, Hughes CW, Oh TH, Fleming J, O'Connor T. Predictors of postoperative respiratory complications in premature infants after inguinal herniorrhaphy. J Pediatr Surg 1993;28:244-247

25. Misra D, Hewitt G, Potts SR, Brown S, Boston VE. Inguinal herniotomy in young infants, with emphasis on premature neonates. J Pediatr Surg 1994;29:1496-1498.

26. Krieger NR, Shochat SJ, McGowan V, Hartman GE. Early hernia repair in the premature infant: long-term follow up. J Pediatr Surg 1994;29:978-81.

27. Parry G, Cline A, Goldmann D. Deciphering Harm Measurement. JAMA 2012;307:2155-2156.

28. Matlow AG, Baker GR, Flintoft V, Cochrane D, Coffey M, Cohen E, Cronin CM, Damignani R, Dube R, Galbraith R, Hartfield D, Newhook LA, Nijssen-Jordan C. Adverse events among children in Canadian hospitals: the Canadian Paediatric Adverse Events Study. CMAJ 2012. DOI:10:1503/cmaj.112153.

29. Leape LL, Brennan TA, Laird N, Lawthers AG, Localio Ar, Barnes BA, Hebert L, Newhouse JP, Weiler PC, Hiatt H. The nature of adverse events in hospitalized patients. Results of the Harvard Medical Practice study II. N Engl J Med 1991;324:377-384.

30. Emanuel L, Berwick D, Conway J, Combes J, Hatlie M, Leape L, Reason J, Schyve P, Vincent C, Walton M. What Exactly Is Patient Safety? In: Henriksen K, Battles JB, Keyes MA, Grady ML, editors. Advances in Patient Safety: New Directions and Alternative Approaches (Vol. 1: Assessment). Rockville (MD): Agency for Healthcare Research and Quality; 2008 Aug.

31. Fan E, Laupacis A, Pronovost PJ, Guyatt GH, Nedham DM. How to use an article about quality improvement. JAMA 2010;304:2279-2287

32. Kuehn BM. FDA considers data on potential risks of anesthesia use in infants, children. JAMA 2011;305:1749-1753.

33. Rappaport B, Mellon RD, Simone A, Woodcock J. Defining safe use of anesthesia in children. N Engl J Med 2011;364:1387-1390.

34. Mellon RD, Simone AF, Rappaport BA. Use of anesthetic agents in neonates and young children. Anesth Analg 2007;104:509-520.

35. Bartels M, Althoff RR, Boomsma DI. Anesthesia and cognitive performance in children: no evidence for a causal relationship. Twin Res Hum Genet 2009;12:246-253.

36. Wilder RT, Flick RP, Sprung J, Katusic SK, Barbaresi WJ, Mickelson C, Gleich SJ, Schroeder DR, Weaver AL, Warner DO. Early exposure to anesthesia and learning disabilities in a population-based birth cohort. Anesthesiology 2009;110:796-804.

37. Ing C, DiMaggio C, Whitehouse A, Hegarty MK, Brady J, von Ungern-Sternberg BS, Davidson A, Wood AJ, Li G, Sun LS. Long-term differences in language and cognitive function after childhood exposure to anesthesia. Pediatrics 2012;130:e476-e485.

38. DiMaggio C, Sun L, Guohua L. Early childhood exposure to anesthesia and risk of developmental and behavioral disorders in a sibling birth cohort. Anesth Analg 2011;

39. Hansen TG, Pedersen JK, Henneberg SW, Pedersen DA, Murray JC, Morton NS, Christensen K. Academic performance in adolescence after inguinal hernia repair in infancy: a nationwide cohort study. Anesthesiology 2011;114:1076-1085.

40. Ramsay JG, Rappaport BA. SmartTots: a multidisciplinary effort to determine anesthetic safety in young children. Anesth Analg 2011;113:963-964.

41. Davidson AJ, McCann ME, Morton N, Myles PS. Anesthesia and outcome after neonatal surgery: the role for randomized trials. Anesthesiology 2008;109:941-944.

42. Blakely ML, Williams R, Dassinger MS, Eubanks JW, III, Fischer P, Huang EY, Paton E, Culbreath B, Hester A, Streck C, Hixson SD, Langham MR, Jr. Early versus interval appendectomy for children with perforated appendicitis. Arch Surg 2011;146:660-665.

43. Myers AL, Williams RF, Giles K, Waters TM, Eubanks JW, III, Hixson SD, Huang Ey, Langham MR Jr., Blakely ML. Hospital cost analysis of a prospective, randomized trial of early versus interval appendectomy for perforated appendicitis in children. J Am Coll Surg 2012;214:427-434.

44. Moss RL, Dimmitt RA, Barnhart DC, Sylvester KG, Brown RL, Powell DM, Islam S, Langer JC, Sato TT, Brandt ML, Lee H, Blakely ML, Lazar EL, Hirschl RB, Kenney BD, Hackam DJ, Zeleterman D, Silverman BL. Laparotomy versus peritoneal drainage for necrotizing enterocolitis and perforation. N Engl J Med. 2006;354:2225-2234.

45. http://clinicaltrials.gov/ct2/show/NCT01029353?term=blakely+m&rank=1. NEST trial.

46. Blakely ML, Tyson JE, Lally KP, McDonald S, Stoll BJ, Stevenson DK, Poole WK, Jobe AH, Wright LL, Higgins RD for NICHD Neonatal Research Network. Pediatrics 2006;117:e680-687.

47. Blakely ML, Lally KP, McDonald S, Brown RL, Barnhart DC, Ricketts RR, Thompson WR, Scherer LR, Klein MD, Letton RW, Chwals WJ, Touloukian RJ, Kurkchubasche AG, Skinner MA, Moss RL, Hilfiker ML for NEC Subcommittee of NICHD Neonatal Research Network. Ann Surg 2005;241:984-989.

48. McCarton CM, Brooks-Gunn J, Wallace IF, Bauer CR, Bennett FC, Bernbaum JC, Broyles RS, Casey PH, McCormick MC, Scott DT, Tyson J, Tonascia J, Meinert CL. Results at age 8 years of early intervention for low-birth-weight premature infants. The Infant Health and Development Program. JAMA 1997;277:126-132.

49. Papile LA, Tyson JE, Stoll BJ, Wright LL, Donovan EF, Bauer CR, Krause-Steinrauf H, Verter J, Korones SB, Lemons JA, Fanaroff AA, Stevenson DK. A multicenter trial of two dexamethasone regimens in ventilator-dependent premature infants. N Engl J Med 1998;338:1112-1118.

50. Tyson JE, Wright LL, Oh W, Kennedy KA, Mele L, Ehrendranz RA, Stoll BJ, Lemons JA, Stevenson DK, Bauer CR, Korones SB, Fanaroff AA. Vitamin A supplementation for extremely-low-birth-weight infants. NICHD Neonatal Research Network. N Engl J Med 1999;340:1962-1968.

51. Ambalavanan N, Tyson JE, Kennedy KA, Hansen NI, Vohr BR, Wright LL, Carlo WA; National Institute of Child Health and Human Development Neonatal Research Network. Pediatrics 2005;115:e249-54.

52. Stark AR, Carlo WA, Tyson JE, Papile LA, Wright LL, Shankaran S, Donovan EF, Oh W, Bauer CR, Saha S, Poole WK, Stoll BJ; NICHD Neonatal Research Network. N Engl J Med 2001;344:95-101.

53. Carlo WA, Stark AR, Wright LL, Tyson JE, Papile LA, Shankaran S, Donovan EF, Oh W, Bauer CR, Saha S, Poole WK, Stoll B. J Pediatr 2002;141:370-374.

54. Broyles RS, Tyson JE, Heyne ET, Heyne RJ, Hickman JF, Swint M, Adams SS, West LA, Pomeroy N, Hicks PJ, Ahn C. Comprehensive follow-up care and life-threatening illnesses among high-risk infants: a randomized controlled trial. JAMA 2000;284:2070-2076.

55. Shankaran S, Laptook AR, Ehrenkranz RA, Tyson JE, McDonald SA, Donovan EF, Fanaroff AA, Poole WK, Wright LL, Higgins RD, Finer NN, Carlo WA, Duara S, Oh W, Cotton CM, Stevenson DK, Stoll BJ, Lemons JA, Guillet R, Jobe AH; NICHD Neonatal Research Network. N Engl J Med 2005;353:1574-1584.

56. Shankaran S, Pappas A, McDonald SA, Vohr BR, Hintz SR, Yolton K, Gustafson KE, Leach TM, Green C, Bara R, Petrie Huitema CM, Ehrenkranz RA, Tyson JE, Das A, Hammond J, Peralta-Carcelen M, Evans PW, Heyne RJ, Wilson-Costello DE, Vaucher YE, Bauer CR, Dusick AM, Adams-Chapman I, Goldstein RF, Guillet R, Papile LA, Higgins RD; Eunice Kennedy Shriver NICHD Neonatal Research Network. Childhood outcomes after hypothermia for neonatal encephalopathy. N Engl J Med 2012;366:2085-2092.

57. Morris BH, Oh W, Tyson JE, Stevenson DK, Phelps DL, O'Shea TM, McDavid GE, Perritt RL, Van Meurs KP, Vohr BR, Grisby C, Yao Q, Pedroza C, Das A, Poole WK, Carlo WA, Duara S, Laptook AR, Salhab WA, Shankaran S, Poindexter BB, Fanaroff AA, Walsh MC, Rasmussen MR, Stoll BJ, Cotten CM, Donovan EF, Ehrenkranz RA, Guillet R, Higgins RD; NICHD Neonatal Research Network. N Engl J Med 2008;359:1885-18896.

58. Hintz SR, Stevenson DK, Yao Q, Wong RJ, Das A, Van Meurs KP, Morris BH, Tyson JE, Oh W, Poole WK, Phelps DL, McDavid GE, Grisby C, Higgins RD; Eunice Kennedy Shriver National Institute of Child Health and Human Development Neonatal Research Network. Is phototherapy exposure associated with better or worse outcomes in 501 to 1000 gram birth weight infants? Acta Paediatr 2011;100:960-965.

59. Tyson JE, Pedroza C, Langer J, Green C, Morris B, Stevenson D, Van Meurs KP, Oh W, Phelps D, O'Shea M, McDavid GE, Grisby C, Higgins R. Does aggressive phototherapy increase mortality while decreasing profound impairment among the smallest and sickest newborns? J Perinatol 2012;32:677-684.

60. Hintz SR, Kendrick DE, Stoll BJ, Vohr BR, Fanaroff AA, Donovan EF, Poole WK, Blakely ML, Wright L, Higgins R; NICHD Neonatal Research Network. Neurodevelopmental and growth outcomes of extremely low birth weight infants after necrotizing enterocolitis. Pediatrics 2005;115:696-703.

61. Stoll BJ, Hansen NI, Adams-Chapman I, Fanaroff AA, Hintz SR, Vohr B, Higgins RD; National Institute of Child Health and Human Development Neonatal Research Network. Neurodevelopmental and growth impairment among extremely low birth weight infants with neonatal infection. JAMA 2004;292:2357-2365.

62. Stephens BE, Bann CM, Watson VE, Sheinkopf SJ, Peralta-Carcelen M, Bodnar A, Yolton K, Goldstein RF, Dusick AM, Wilson-Costello DE, Acarregui MJ, Pappas A, Adams-Chapman I, McGowan EC, Heyne RJ, Hintz SR, Ehrenkranz RA, Fuller J, Das A, Higgins RD, Vohr BR; for the Eunice Kennedy Shriver National Institute of Child Health and Human Development Neonatal Research Network. Screening for autism spectrum disorders in extremely preterm infants. J Dev Behav Pediatr 2012;33:535-541.

63. Etchegaray JM, Gallagher TH, Bell SK, Dunlap B, Thomas EJ. Error disclosure: a new domain for safety culture assessment. BMJ Qual Saf 2012;21:594-599.

64. Profit J, Etchegaray J, Petersen LA, Sexton JB, Hysong SJ, Mei M, Thomas EJ. Neonatal intensive care unit safety culture varies widely. Arch Dis Child Fetal Neonatal Ed. 2012;97:F120-126.
65. Gawande AA, Thomas EJ, Zinner MJ, Brennan TA. The incidence and nature of surgical adverse events in Colorado and Utah in 1992. Surgery 1999;126:66-75.
66. Tyson, J. Scientific evaluation of new infant formulas for preterm infants. An expanded view. FDA White Paper, 2002. http://www.fda.gov/ohrms/dockets/ac/01/briefing/3903b1.htm.
67. Tyson JE, et al. An evaluation of the quality of therapeutic studies in perinatal medicine. With Editorial. J Pediatr 102:10, 1983. See also Early Hum Devel 10:175, 1985, JAMA 272:166,1994; & BMJ 327: 947, 2003.
68. Kao LS, Tyson JE, Blakely ML, Lally KP. Clinical research methodology: introduction to randomized trials. J Am Coll Surg 2008;206:361-369.
69. Tyson JE, Knudson PL. Views of neonatologists and parents on consent for clinical trials. Lancet 356:2026-2027, 2000
70. Rogers CG, Tyson JE, Kennedy KA, Broyles RS, Hickman JF. Conventional consent with opting in versus simplified consent with opting out: an exploratory trial for studies that do not increase patient risk. J Pediatr 132:606-611, 1998.
71. Sinclair JC. Weighing risks and benefits in treating the individual patient. Clin Perinatol. 2003 Jun;30(2):251-68.
72. Kent DM. Hayward RA. Limitations of applying summary results to clinical trials to individual patients; the need for risk stratification. JAMA 2007; 298:1209-15
73. Hayward RA, Kent DM, Vijan S, Hofer TP. Reporting clinical trial results to inform providers, payers, and consumers. Health Aff (Millwood). 2005 Nov-Dec;24(6):1571-81
74. Sun X, Briel M, Busse JW. Credibility of claims of subgroup effects in randomised controlled trials: systematic review. BMJ. 2012;344:e1553.
75. Garcia EF, Nguyen H, Duan N, et al. Assessing heterogeneity of treatment effects: Are authors misinterpreting their results? Health Serv Res 2010;45:283-301.
76. Kraemer HC, Frank E, Kupfer DJ. Moderators of treatment outcomes: clinical, research, and policy importance. JAMA 2006;296:1286-1289.
77. Kraemer HC, Frank E. Evaluation of comparative treatment trials: assessing clinical benefits and risks for patients, rather than statistical effects on measures. JAMA 2010; 304:683-684.
78. Wijeysundera DN, Austin PC, Hux JE, Beattie WS, Laupacis A. Bayesian statistical inference enhances the interpretation of contemporary randomized controlled trials. J Clin Epidemiol. 2009;62(1):13-21.e5.
79. Lilford RJ, Thornton JG, Braunholtz D. Clinical trials and rare diseases: a way out of a conundrum. BMJ 1995;311(7020):1621-1625.
80. Spiegelhalter DJ, Myles JP, Jones DR, Abrams, KR. An introduction to Bayesian methods in health technology assessment. BMJ 1999;319:508-512.
81. Dixon DO, Simon R. Bayesian Subset Analysis. Biometrics 1991;47(3):871–881.
82. Dorresteijn JA, Visseren FL, Ridker PM, Wassink AM, Paynter NP, Steyerberg EW, van der Graaf Y, Cook NR. Estimating treatment effects for individual patients based on the results of randomised clinical trials. BMJ. 2011 Oct 3;343:d5888.
83. Tyson JE, Parikh NA, Langer J, Green C, Higgins RD; National Institute of Child Health and Human Development Neonatal Research Network. Intensive care for extreme prematurity—moving beyond gestational age. N Engl J Med 2008;358:1672-1681.
84. Greene SM, Geiger, AM. A review finds that multicenter studies face substantial challenges but strategies exist to achieve Institutional Review Board approval. Journal of Clinical Epidemiology 2006;59:784–790.
85. Greene S, Geiger A, Harris E. Impact of IRB requirements on a multicenter survey of prophylactic mastectomy outcomes. Annals of Epidemiology 2006;16: 275–278.
86. Helfand BT, Mongiu AK, Roehrborn CG, Donnell RF, Bruskewitz R, Kaplan SA, Kusek JW, Coombs L, McVary KT; MIST Investigators. Variation in institutional review board responses to a standard protocol for a multicenter randomized, controlled surgical trial. Journal of Urology 2009;181:2674–2679.
87. Sherwood ML, Buchinsky FJ, Quigley MR, Donfack, J, Choi SS, Conley SF, Derkay CS, Myer CM III, Ehrilich GD, Post JC. Unique challenges of obtaining regulatory approval for a multicenter protocol to study the genetics of RRP and suggested remedies. Otolaryngology–Head and Neck Surgery 2006; 135:189–196.

88. Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap) - a metadata-driven methodology and workflow process for providing translational research informatics support. J Biomed Inform 2009;42:377-381.
89. Zwarenstein M, Treweek S, Gagnier JJ, Altman DG, Tunis S, Haynes B, Oxman AD, Moher D for the CONSORT and Pragmatic Trials in Healthcare (Practihc) groups. BMJ 2008;337:1-8.
90. Bender BG, Ellison MC, Gleason M, Murphy JR, Sundstrom DA, Szefler SJ. Minimizing attrition in a long-term clinical trial of pediatric asthma. Ann Allergy Asthma Immunol 2003;91:168-176.
91. Oken A, Rasmussen MD, Slagle JM, Jain S, Kuykendall T, Ordonez N, Weinger MB. A facilitated survey instrument captures significantly more anesthesia events than does traditional voluntary event reporting. Anesthesiology 2007;107:909-922.
92. Forrey RA, Pedersen CA, Schneider PJ. Interrater agreement with a standard scheme for classifying medication errors. Am J Health-Syst Pharm. 2007;64:175-181.
93. Duthie E, Favreau B, Ruperto A, Mannion J, Flink E, Leslie R. Quantitative and qualitative analysis of medication errors: the New York experience. In: Henriksen K, Battles JB, Marks ES, Lewin DI, editors. Advances in Patient Safety: From Research to Implementation (Volume 1: Research Findings). Rockville (MD): Agency for Healthcare Research and Quality (US); 2005.
94. National Coordinating Council for Medication Error Reporting and Prevention. NCC MERP index for categorizing medication errors. http://www.nccmerp.org/pdf/indexColor2001-06-12.pdf (accessed September 16, 2012).
95. Vohr BR, Wright LL, Dusick AM, Perritt R, Poole WK, Tyson JE, Steichen JJ, Bauer CR, Wilson-Costello DE, Mayes LC. Center differences and outcomes of extremely low birth weight infants. Pediatrics 2004;113:781-9.
96. Bayley N. Bayley Scales of Infant Development. Third Edition. San Antonio, TX: The Psychological Corporation; 2006.
97. Rosenbaum P, Paneth N, Leviton A, Goldstein M, Bax M. The Definition and Classification of Cerebral Palsy. Dev Med Child Neurol 2007; 49(Suppl. 109): 8–14.
98. Palisano R, Rosenbaum P, Walter S, Russell D, Wood E, Galuppi B. Development and reliability of a system to classify gross motor function in children with cerebral palsy. Dev Med Child Neurol 1997;39:214-23.
99. Dixon DO, Simon R. Bayesian Subset Analysis. Biometrics 1991;47:871-881.
100. Gelman A, Meng XL, and Stern HS. Posterior Predictive Assessment of Model Fitness Via Realized Discrepancies. Statistica Sinica 1996;6:733-807.
101. Spiegelhalter D, Pearson M, Short I. Visualizing uncertainty about the future. Science 2011;333:1393-1400.
102. Atwood JR, Giordano L, Vargas P, Blackwell GG, Earnest DL, Meyskens F, Alberts D. Adherence enhancers in pill-related clinical trials: a health behavior in cancer prevention model-based approach. Patient Educ Couns 1996;28:15-23.
103. Schulz KF, Grimes DA. Sample size calculations in randomized trials: mandatory and mystical. Lancet. 2005 Apr 9-15;365(9467):1348-53
104. Gollenberg AL, Lynch CD, Jackson LW, McGuinness BM, Msall ME. Concurrent validity of the parent-completed Ages and Stages Questionnaires, 2nd Ed. With the Bayley Scales of Infant Development II in a low-risk sample. Child Care Health Dev. 2010;36:485-90.
105. Consolodated Standards of Reporting Trials (CONSORT) Transparent Reporting of Trials:2010 Statement. . http://www.consort-statement.org/consort-2010 (accessed December 18, 2014).

2. Serious Adverse Events And Definitions For HIP Trial To Be Recorded During The Study Period.

Serious adverse event*	Definition
Death	Death from any cause identified from medical records, vital statistics records, or maternal report (verified from other sources whenever possible)
Use of extracorporeal membrane oxygenation (ECMO)	ECMO cannulation performed (for any cause) whether or not ECMO was successfully initiated
Cardiac arrest	Heart rate of zero documented in medical records
Cardiopulmonary Resuscitation (CPR)	Administration of ventilation and cardiac massage (with or without medications or other interventions) documented in medical records.
Apnea treated with respiratory support	One or more episodes of apnea that as noted in the nursing or physician notes was treated with an increased FiO_2 AND an increase in respiratory support, e.g. use of bag and mask ventilation or bag to tube ventilation, or the initiation or increase in mechanical ventilation
Hypotension treated with vasoactive drugs	Administration of vasopressor or others vasoactive medications to raise blood pressure or improve circulation to treat infants whose blood pressure falls below age specific standards. Fluid bolus alone or continued administration of vasoactive drugs at the same dose as before surgery does not indicate a SAE for HIP trial purposes.
Persistent bradycardia that received treatment	Low heart rate treated with pharmacologic intervention
Local anesthetic toxicity	Any adverse event presumed to be due to local anesthetic as ascertained from anesthesia or PACU[**] pediatric ICU, or NICU records
Regional anesthetic toxicity	Any adverse event presumed to be due to regional anesthetic medication should be recorded; ascertained from anesthesia and PACU, pediatric ICU, or NICU records
Prolonged intubation[++]	Patient intubated and receiving mechanical ventilation>48 hours from completion of IH repair procedure. Only for infants not requiring mechanical ventilation prior to surgery
Unplanned reintubation[++]	Reintubation that was not planned to be performed
Intraoperative unplanned extubation	Inadvertent extubation during surgery. Record whether the patient requires reintubation or not.
IH incarceration	IH that is nonreducible AND necessitates an urgent IH repair; OR narcotic pain medication or sedation is given to facilitate IH reduction; OR reduction requires the pediatric surgery team, which documents IH incarceration in the patient’s record
IH strangulation	An incarcerated IH in which intra hernia contents are ischemic or otherwise damaged from the incarceration, as identified at operation
IH recurrence	Return of the IH during the study period, as documented on physical exam or imaging AND documented in the patient’s record
Reoperation for IH or related to initial operation	A second, or subsequent, operation, also focused on an IH (either the same side as originally, or the contralateral side); OR

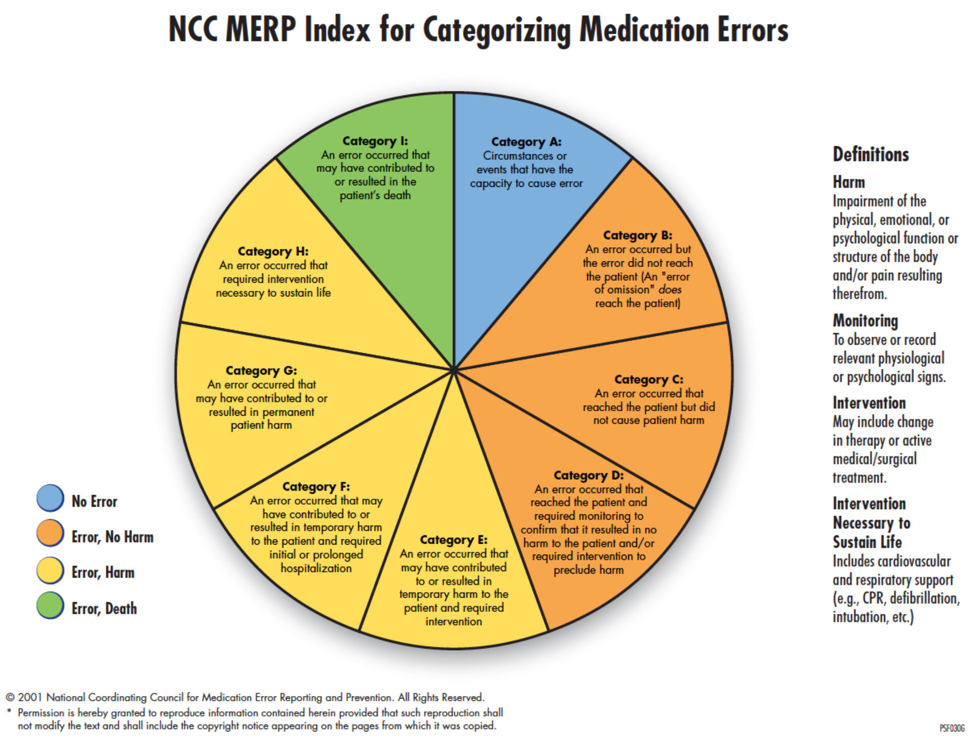
	a subsequent operation that was indicated due to a SAE from the IH repair (e.g. laparotomy for bowel injury during initial IH repair)
Injury to adjacent structure / organ	Injury thought to result from surgery that was identified the IH repair operation (and recorded in the operative report) or subsequently
Wound disruption (deep)++	Dehiscence of the surgical wound, including the deep (external oblique) fascia
Surgical site infection (deep)++	Infection occurring within 30 days after IH repair, which involves deep soft tissues (e.g. fascia and muscle) AND at least ONE of the following: purulent drainage from deep incision; deep incision dehisces or is deliberately opened by provider; abscess is documented in medical record; deep SSI is recorded by surgeon or other attending physician
Other SAE that requires intervention to prevent or address patient harm	Any serious adverse event that is recorded in the patient record that requires specific intervention to address or prevent harm

*The period of ascertainment is from enrollment through 10 months after enrollment.

**PACU = post anesthesia care unit (aka. Recovery room)

++Adapted from existing definition used by American College of Surgeons Pediatric National Surgical Quality Improvement Project (ACS NSQIP). Dr. Blakely (HIP Trial PI) serves on the national data definitions committee for NSQIP.

3. National Coordinating Council for Medication Error Reporting and Prevention. NCC MERP index for categorizing medication errors.



Protocol changes:

Version 2 (5/15/13) – updated title page and data elements; removed “Study” from header; added “at selected sites with well established follow up programs” to Secondary Aims 1; added “Daniel Roke, MD” to Table 5; added “Institutional Board Review” and “Trial Commencement” to end of protocol.

Version 3 (12/30/14) – updated title page to include Andrea Krzyzaniak as study coordinator.

Version 4 (2/13/15) – updated Table 2 Participating Clinical Sites and Site Surgery Investigators and open the neurodevelopmental follow-up portion to sites outside of the Neonatal Research Network (NRN). The neurodevelopmental follow-up will be available to the first 200 participants at sites (both NRN and non-NRN) with programs capable of administering a neurologic exam and BSID-III with current resources.

Version 5 (4/27/16) – updated title page to current terminology from “post conceptual age” to “post menstrual age). Updated study investigators to replace Drs. Steven Haynes and Nathalie Maitre to Dr. Joern-Hendrick Weitkamp. Updated patient recruitment methods to include SUBJECT LOCATOR. Updated Table 2 Participating Clinical Sites and Site Surgery Investigators. Provided additional clarification on Exclusion Criteria.

Version 6 (12/20/18) updated outcome adjudication committee personnel. Updated Table 2 Participating Clinical Sites and Site Surgery Investigators. Deleted Appendix I as all database fields are outlined in complete detail in the study’s Manual of Operations (MOP).

Version 7 (11/27/2019) due to slower than planned enrollment, a safety and efficacy analysis will be completed in December 2019 for approximately 200 infants instead of two interim safety analyses at 33% and 66% enrollment. Also, a final Bayesian analysis of the primary outcome and major secondary outcomes will be completed in addition to the planned final frequentist analysis.